

HealthNest

CS673 Software Engineering | Team 2 | Iteration 2 Presentation

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What is HealthNest?

HealthNest is an AI-powered healthcare platform designed to connect patients, providers, and care teams in one place. Patients and providers can communicate, schedule appointments, access lab results, and interact with AI-powered tools.

The technology stack consists of a React frontend, Python FastAPI backend, Supabase database, and OpenAI API integration.

Patients

Booking appointments, view lab results, interact with Pulse AI

Providers

Upload results, manage schedule, review patient alerts

Pulse AI

Streaming AI assistant with appointment and lab result skills

Iteration 2 Goals



Pulse AI

Patient-facing streaming AI with appointment and lab result skill cards.

Provider-facing DFA visit overview summary skill added for clinical workflow support.



Biometric Authentication

Passkey sign-in fully working for both Patient and Provider roles. Separate emails are required per role.



Appointment Scheduler

Full patient booking, reschedule, and cancellation workflow. Backend-connected.



Lab Results

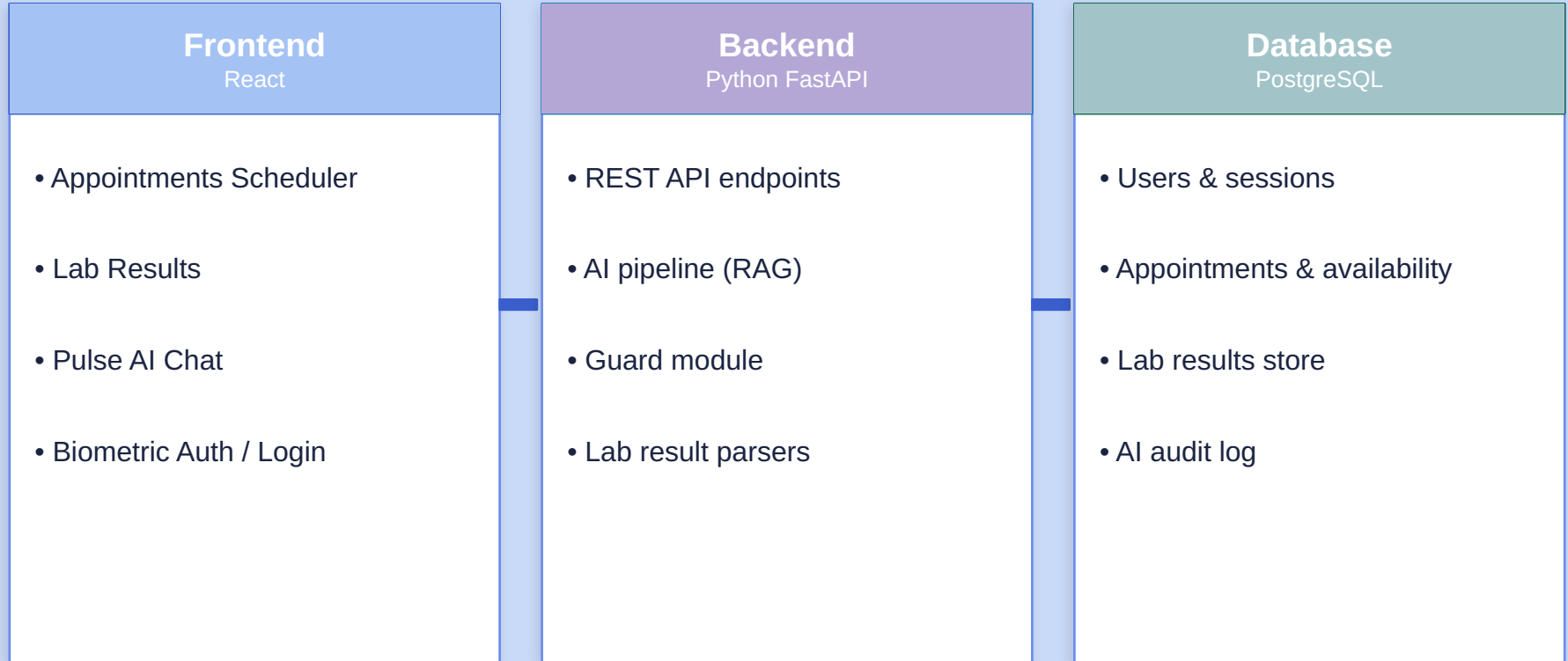
Provider upload, review, and release workflow. Patient view includes flagged values.



Unit Testing Framework

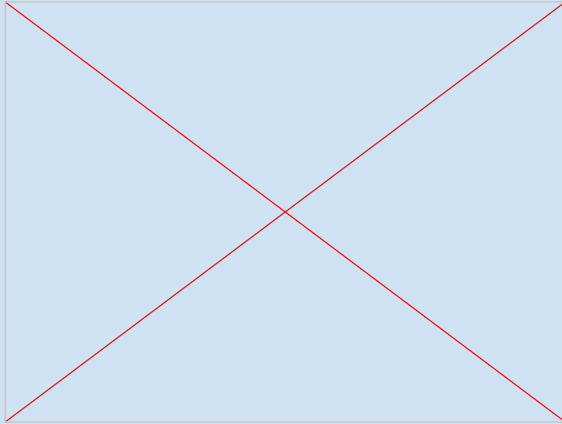
191 Jest tests and 105 backend pytest tests across 38 files.

System Architecture



Pulse AI & DFA Visit Overview

Patient-Facing Pulse AI



- Able to schedule appointments
- Able to view lab reports
- Able to further explain doctor directives
- Helps patients understand available actions

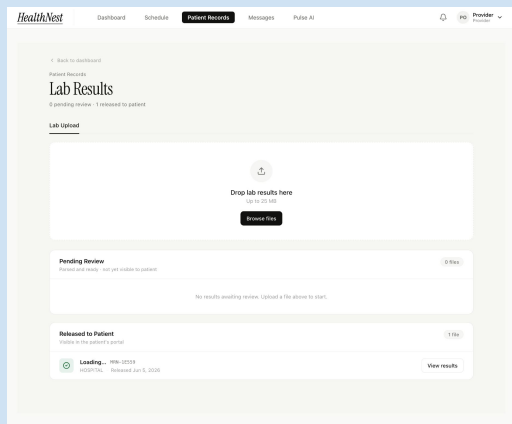
Provider-Facing Visit-Overview

The dashboard displays the following sections:

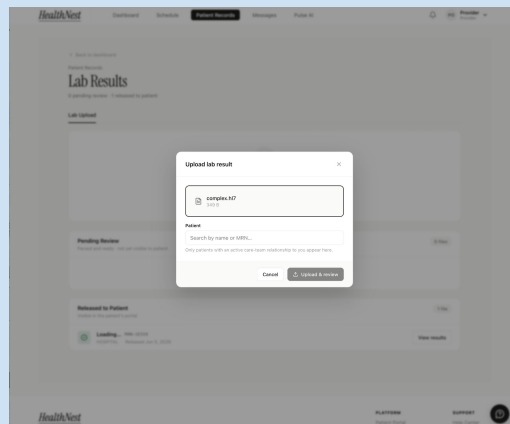
- Today's Patients:** 0 scheduled, 0 seen - 0 pending
- Unsigned Encounters:** 0 notes
- Clinical Alerts:** 0 active
- AI Pre-Visit Summaries:** Generated from records, labs, and prior notes. 0 upcoming.
- Patient Alerts:** 0 active
- Today's Schedule:** Jun 5

- Added DFA visit overview summary skill
- Summarizes visit-related information for providers
- Supports provider workflow inside the doctor dashboard
- Expands AI support beyond patient chat

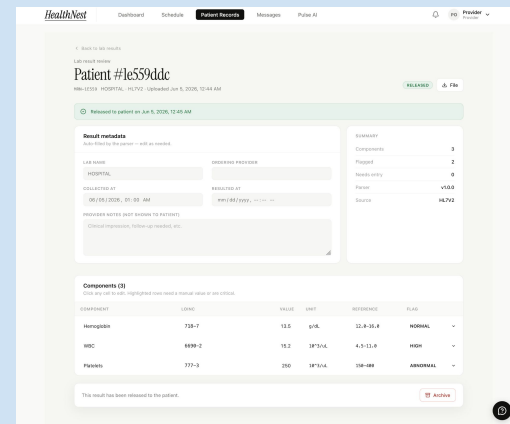
Lab Results



Upload File

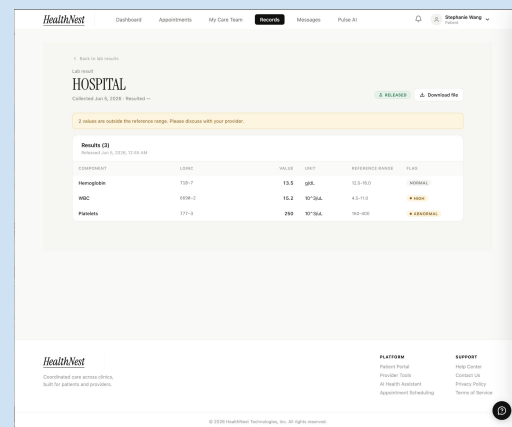


Review & Release



Full Lifecycle: provider upload → parse → review → release → patient view

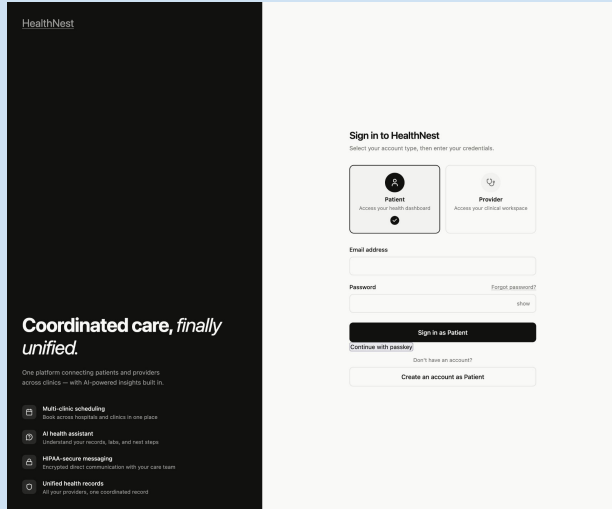
- Supports HL7, JSON, and XML uploads
- Providers review parsed results before release
- Patient view highlights flagged values



Patient View

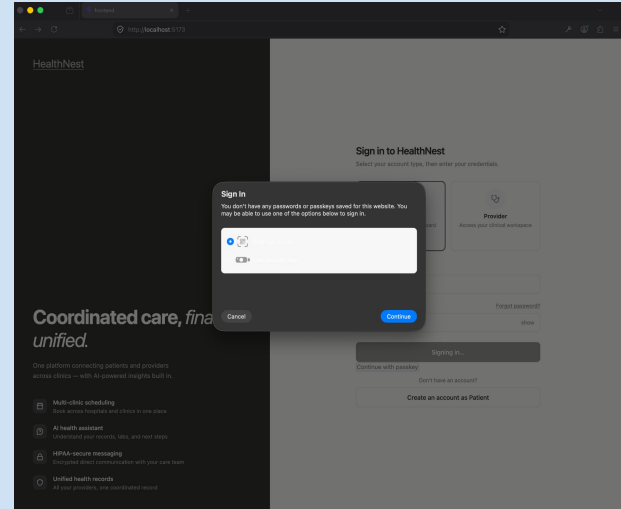
Biometric Authentication

What changed in Iteration 2



- Passkey sign-in added for patient and provider roles
- Supports role-based authentication flow
- Separate email accounts are required per role
- Routes users to the correct dashboard after login

Why it matters

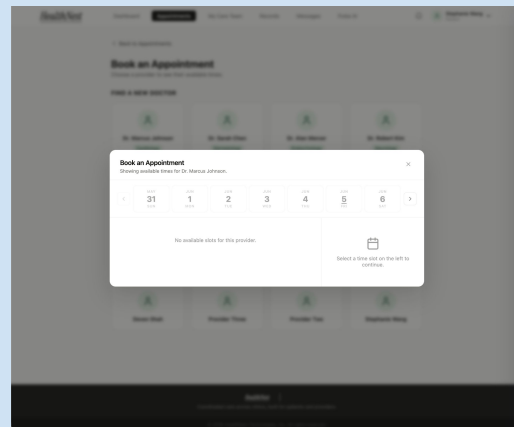
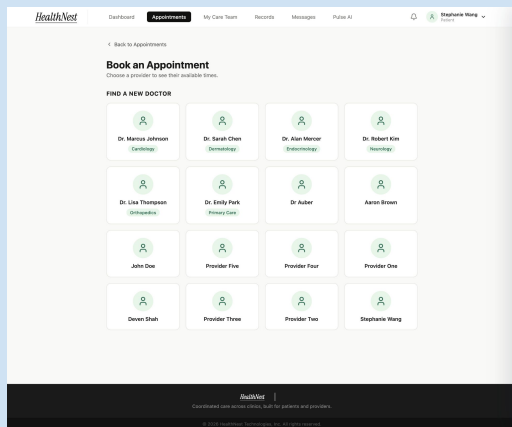
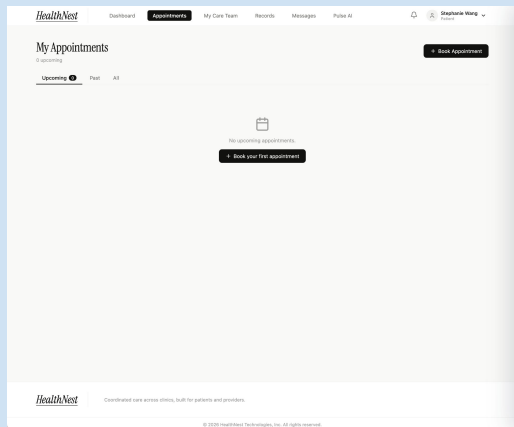


- Improves sign-in security
- Reduces password-only dependency
- Keeps patient and provider access separate
- Builds on the Iteration 1 login/sign-up flow

Appointment Scheduler

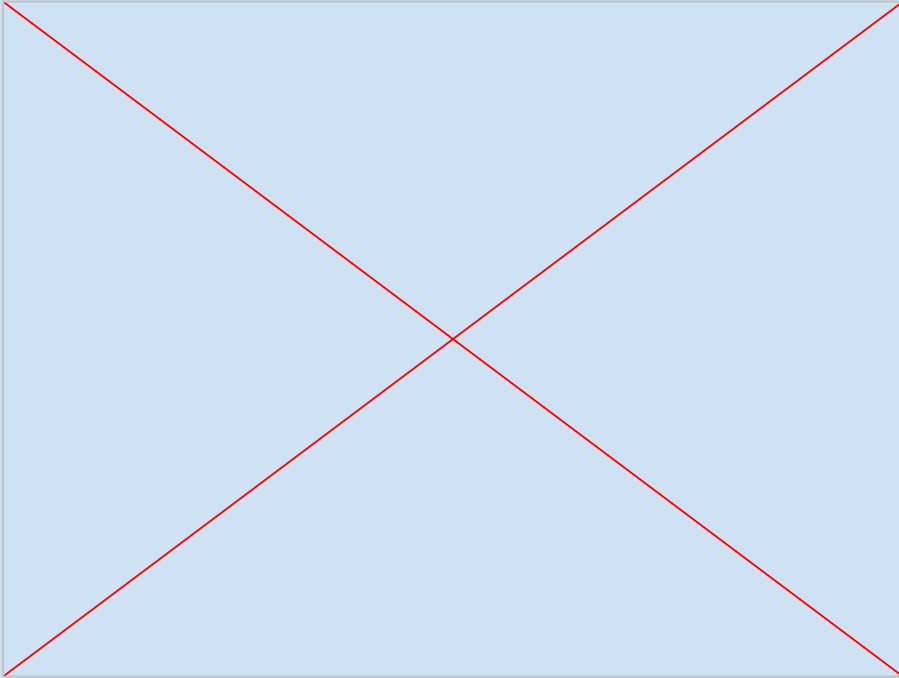
Patient Booking Workflow

- Backend-connected appointment workflow
- Patients can book, reschedule, and cancel appointments
- Provider availability is used to display open time slots
- Appointment data model was refactored around provider and availability relationships



Secure Messaging

Real-Time Secure Messaging



- Real-time secure messaging between patients and providers
- Conversation-based interface with contact lists
- Role-aware messaging workflows for both patient and provider dashboards
- Realtime message delivery using Supabase Realtime subscriptions

Testing & QA Coverage

Active	Manual / Acceptance	14 manual test cases tracked: TC001–TC014. Login and lab results passed; appointment booking found one issue.
Active	System Testing	Local Docker workflows tested: sign-in → appointment workflow → lab upload → release → patient view
Active	Unit Testing — Frontend (Jest)	191 tests across 25 test files.
Active	Unit Testing — Backend (pytest)	105 tests across 13 test files.
Active	CI/CD Safety Checks	Jira validation, frontend lint/build, Docker build, and Docker Compose validation on PRs.
Partial	Integration Testing	Auth and lab flows verified; appointment flow partially blocked by TC008 availability endpoint issue.
Planned	Automated Regression Suite	Jest + pytest tests exist locally but are not yet required CI/CD gates. Planned for Iteration 3.

CI/CD & Testing Metrics

1

Validate Jira Key

Branch name must match feature/SCRUM-XX-description.
Commit message must contain a Jira ticket ID.

2

Frontend Build & Lint

npm ci → ESLint on all .js/.jsx files → npm run build.
PRs blocked if any step fails.

3

Docker Build Check

Frontend image + backend image built from scratch.
docker-compose.yml is also validated.

4

Release Build

Runs on release/* branches only.
Rebuilds both images and re-validates compose for release readiness.

Iteration 2 Stats

191

Frontend Jest tests — 191 passed / 0 failed

105

Backend pytest tests — 105 passed / 0 failed

14

Manual test cases —
11 passed / 1 Failed / 2 Blocked

Sprint Backlog – Iteration 2

✓ Completed for Iteration 2 Scope

 In Progress

SCRUM-7 / 56

User Login + Biometric Auth

SCRUM-5

Appointment Scheduler

SCRUM-61

DFA Visit-Overview Skill

SCRUM-39

Pulse AI (Patient-facing)

SCRUM-55

AI Escalation Path

SCRUM-54

Secure Messaging

SCRUM-27

Doctor-Facing AI Assistant

SCRUM-57

E-Prescribing and Refill Workflow

What's Next — Iteration 3

Feature Development

- Complete Secure Messaging frontend/backend flow
- Continue DFA Visit-Overview Skill integration
- Resolve appointment availability issue from TC008
- Strengthen Role-Based Access Control at API endpoints

Testing & Quality

- Add Jest + pytest as required CI/CD gates
- Expand integration testing for appointment and lab workflows
- Add backend testing for biometric/passkey authentication
- Add AI/RAG quality testing beyond functional tests

Data & Infrastructure

- Connect Patient and Provider Dashboards to live backend data
- Add demo test accounts to reduce email verification friction
- Improve README setup instructions for Docker/local testing
- Document HIPAA/security risks in STD: audit trail, access control, input sanitization

Thank you!